Fact: $f: X \to Y$

TFAE: (a) f is a bijection,

(b) $\exists\, g: Y \to X$ with the following property:

$\forall x \in X,\ y \in Y$

$$\begin{tikzcd} x \arrow[r, mapsto, "f"] & f(x) \arrow[d, mapsto, "g"] \\ & g(f(x)) = x \arrow[ul] \end{tikzcd} \qquad \begin{tikzcd} y \arrow[r, mapsto, "g"] & g(y) \arrow[d, mapsto, "f"] \\ & f(g(y)) = y \arrow[ul] \end{tikzcd}$$

i.e.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr] & Y \arrow[d, "g"] \\ & X \end{tikzcd} \qquad \& \qquad \begin{tikzcd} Y \arrow[r, "g"] \arrow[dr] & X \arrow[d, "f"] \\ & Y \end{tikzcd}$$

$id_X = g \circ f$ $\qquad$ $id_Y = f \circ g$

Pf of Fact | (b) $\Rightarrow$ (a)

(ONTO) $y \in Y$; then $y = f(g(y))$

(ONE-TO-ONE) $x_1, x_2 \in X$ s.t. $f(x_1) = f(x_2)$

$$\begin{tikzcd} f(x_1) \arrow[d, mapsto, "g"] & f(x_2) \arrow[d, mapsto, "g"] \\ g(f(x_1)) \arrow[r, phantom, "="] & g(f(x_2)) \end{tikzcd}$$

$$x_1 \overset{\checkmark}{=} x_2$$

(c) $\Rightarrow$ (b)

f is one-to-one & onto

Define $g: Y \longrightarrow X$ by the rule:

$g(y) =$ the unique x s.t. $f(x) = y$

Then we have $\forall y \in Y$

$$\underbrace{f(g(y)) = y}$$ BY DEFINITION!

But what is $g(f(x))$? (for $x \in X$)

For this, note:

$$\underbrace{f(g(f(x)))}_{\text{(why?) Take } y = f(x)} = \underbrace{f(x)} \xRightarrow[\text{is one-to-one}]{f} \underbrace{g(f(x)) = x}_{\substack{\Uparrow \\ X}}$$

Def$^{\underline{n}}$ In this situation,
f & g are inverses to each other. ; $g = f^{-1}$ (Notation)

We will also say that f is invertible.

Example:

$$\left.\begin{aligned} f : \mathbb{R}_{\geq 0} &\longrightarrow \mathbb{R}_{\geq 0} \\ x &\longmapsto x^2 \end{aligned}\right\}$$ ONE-TO-ONE & ONTO

The inverse is

$$\begin{aligned} g : \mathbb{R}_{\geq 0} &\longrightarrow \mathbb{R}_{\geq 0} \\ y &\longmapsto \sqrt{y} \end{aligned}$$

Linear transformations

Def$^{\underline{n}}$: V, W: F-vector spaces

An F-linear transformation (or just linear transformation if F is clear from context) is a function

$$T : V \longrightarrow W$$

s.t.

(a) (Compatibility with $+$)

$v_1, v_2 \in V$

$$\underbrace{T(v_1 + v_2)}_{\in V} = \underbrace{T(v_1) + T(v_2)}_{\in W}$$

(b) (Compatibility with $\cdot$)

$\forall a \in F,\ v \in V$

$$\underbrace{T(\underbrace{a\cdot v}_{\in V})}_{} = \underbrace{a\cdot T(v)}_{\in W}$$

Rmk 1 (Compatibility w. 0)

- $T(0_V) = T(0\cdot v)$
$= 0\cdot T(v)$
$= 0_W \in W$

- $T(0_V + 0_V) = T(0_V) + T(0_V)$

$$\begin{array}{ccc} T(0_V + 0_V) & = & T(0_V)+T(0_V) \\ \| & & \| \\ T(0_V) & = & T(0_V)+T(0_V) \end{array}$$

$\{ - T(0_V)$

$0_W = T(0_V) \in W$

(2) (Compatibility with F-linear combos)

$$T\left(\sum_{i=1}^{n} a_i v_i\right) = T(a_1v_1 + \dots + a_nv_n)$$

$$(\forall a_i \in F,\ v_i \in V) \qquad = a_1T(v_1) + \dots + a_nT(v_n)$$

$$= \sum_{i=1}^{n} a_i T(v_i)$$

Examples:

(1)
$$\begin{array}{ccc} V & \longrightarrow & W \\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ v & \longmapsto & 0 \end{array}$$

The zero or trivial linear transformation $\left.\right\}$ $\cdot\ 0+0=0$ $\cdot\ a\cdot 0=0$

(2) $id_V : V \longrightarrow V$

$v \longmapsto v$

The identity linear transformation

(3) $D : \mathrm{Poly}(\mathbb{R}) \longrightarrow \mathrm{Poly}(\mathbb{R})$

$$f(x) \longmapsto D(f(x)) = f'(x)$$

$$\sum_{i=0}^{n} a_i x^i \longmapsto \sum_{i=0}^{n} i a_i x^{i-1}$$

Check:

- $$D(f_1(x) + f_2(x)) \overset{?}{=} D(f_1(x)) + D(f_2(x))$$
  $$(f_1 + f_2)'(x) \overset{\checkmark}{=} f_1'(x) + f_2'(x)$$

- $$D(a \cdot f(x)) \overset{?}{=} a \cdot D(f(x))$$
  $$(a \cdot f)'(x) \overset{\checkmark}{=} a \cdot f'(x)$$

Rmk: The formula makes sense for any field, not just $\mathbb{R}$

Caution:

$F = \mathbb{Z}/p\mathbb{Z}$

$$D(x^p) = p x^{p-1} = 0!$$

(4) $T : F^2 \longrightarrow F^2$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \longmapsto \begin{bmatrix} p a_1 + q a_2 \\ r a_1 + s a_2 \end{bmatrix}$$

$p, q, r, s \in F$

?

$$\cdot\ T\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}\right) \overset{?}{=} T\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) + T\left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}\right)$$

$$= T\left(\begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}\right) = \begin{bmatrix} p(a_1+b_1) + q(a_2+b_2) \\ r(a_1+b_1) + s(a_2+b_2) \end{bmatrix}$$

$$T\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) + T\left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}\right) = \begin{bmatrix} pa_1 + qa_2 \\ ra_1 + sa_2 \end{bmatrix} + \begin{bmatrix} pb_1 + qb_2 \\ rb_1 + sb_2 \end{bmatrix} = \begin{bmatrix} pa_1 + qa_2 + pb_1 + qb_2 \\ ra_1 + sa_2 + rb_1 + sb_2 \end{bmatrix}$$

$$\begin{bmatrix} p(a_1+b_1) + q(a_2+b_2) \\ r(a_1+b_1) + s(a_2+b_2) \end{bmatrix} \overset{\checkmark}{=} \begin{bmatrix} pa_1 + qa_2 + pb_1 + qb_2 \\ ra_1 + sa_2 + rb_1 + sb_2 \end{bmatrix}$$

$$\cdot\ T\left(c \cdot \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) \overset{?}{=} c \cdot T\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right)$$

$$T\left(c \cdot \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} ca_1 \\ ca_2 \end{bmatrix}\right)$$

$$\checkmark$$

Def^{n} $T: V \to W$: linear transformation

(2) T is invertible if it is bijective

(3) The kernel of T is the subset

$$\ker T = \{ v \in V : T(v) = 0 \in W \}$$

$$\subseteq V$$

(4) The image of T is the subset

$$\operatorname{im} T = \{ T(v) : v \in V \}$$

$$\subseteq W$$